Diffusion models for trajectory planning

Nick Bermbach

I. Introduction

- Trajectory planning with multiple feasible routes, goals, and obstacle constraints
- Classical trajectory optimization versus learning a distribution of plausible trajectories
- Diffuser's architecture and conditioning, diffusion-based control policies, and a 2D planning study

II. Problem formulation and diffusion basics

A. Trajectory planning

- State-action trajectories, system dynamics, and planning horizon; classical trajectory optimization as background [5]
- Start and goal conditions; collision avoidance, path length, and smoothness
- Classical optimization as a comparison: CMA-ES on a parameterized trajectory [2]

B. Denoising diffusion models

- Forward noising, learned reverse process, and noise-prediction objective (DDPM) [3]
- Distinction between physical trajectory time and diffusion timesteps
- Iterative sampling cost; DDIM as an optional accelerated sampler [7]

III. Diffuser: learning and conditioning trajectory distributions

A. Representation, architecture, and training

- Joint state-action trajectories; temporal U-Net and parallel denoising across the planning horizon [4]
- Learning from offline trajectory data; multimodality and dependence on data coverage

B. Goal conditioning and planning objectives

- Inpainting: fixing the current state and specified goals during iterative denoising
- Reward guidance using a separately learned return predictor; modifying the denoising distribution
- Planning as inference: $\widetilde{p}_\theta(\tau) \propto p_\theta(\tau) h(\tau)$ [6]
- Maze2D navigation with start-goal inpainting, in contrast to reward-guided locomotion

C. Constraints and limitations

- Obstacle avoidance learned from trajectory data versus explicit test-time cost guidance
- Inpainting of specified values versus soft guidance of trajectory costs
- No general guarantees for collision avoidance or consistency with system dynamics

IV. From Diffuser to diffusion policies

A. Diffusion Policy

- Observation-conditioned generation of action sequences instead of joint state-action trajectories [1]
- Behavior cloning from demonstrations; visual conditioning with temporal CNN or transformer
- Receding-horizon execution and the trade-off between action consistency and responsiveness

B. One-Step Diffusion Policy

- Distillation of a pretrained diffusion-policy teacher into a single-pass action generator [8]
- Distribution matching using reverse-KL/score-difference objectives
- One-step inference, not one-step training; teacher and distillation costs remain

C. Comparison of the three methods

- Generated variables, conditioning inputs, training data, and inference procedures
- Planning versus policy learning; use of receding-horizon execution in Diffuser and Diffusion Policy
- Flexibility, multimodality, runtime, and applicability to the 2D experiment

V. Experimental study: 2D trajectory planning

A. Environment and training data

- Point robot with single-integrator dynamics, static obstacles, and start-goal queries
- Diverse, collision-free demonstration trajectories; disjoint train/validation/test queries
- Comparison with the CMA-ES seminar on a common benchmark

B. Methods and evaluation

- Simplified Diffuser-style model jointly generating states and actions, with start-goal inpainting
- Inpainting-only baseline versus additional obstacle-cost guidance; CMA-ES on an equivalent task
- Feasibility (collisions, dynamics, speed bound), plan quality, route diversity, and inference runtime

C. Ablations and failure analysis

- Main ablation: contribution of obstacle-cost guidance beyond the learned trajectory prior
- Role of training-data coverage; representative collision and dynamics failures
- Limits of conclusions drawn from a simple 2D setting

VI. Discussion and conclusion

- Comparison of the three papers and the completed 2D experiment
- Trade-offs among data requirements, flexibility, feasibility, solution quality, and computational cost
- Limitations of learned constraints and iterative sampling; open questions for real-world robot control

References

[1] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44(10-11):1684–1704, 2025.

[2] Nikolaus Hansen. The CMA evolution strategy: A tutorial, 2016. Revised version, 2023.

[3] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[4] Michael Janner, Yilun Du, Joshua B. Tenenbaum, and Sergey Levine. Planning with diffusion for flexible behavior synthesis. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 9902–9915. PMLR, 2022.

[5] Matthew Kelly. An introduction to trajectory optimization: How to do your own direct collocation. *SIAM Review*, 59(4):849–904, 2017.

[6] Sergey Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review, 2018.

[7] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.

[8] Zhendong Wang, Zhaoshuo Li, Ajay Mandlekar, Zhenjia Xu, Jiaojiao Fan, Yashraj Narang, Linxi Fan, Yuke Zhu, Yogesh Balaji, Mingyuan Zhou, Ming-Yu Liu, and Yu Zeng. One-step diffusion policy: Fast visuomotor policies via diffusion distillation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 63399–63416. PMLR, 2025.